
MODULE-WISE FUNCTIONALITY & WORKFLOW

AI-HRMS

Comprehensive module-wise functionality document with configured workflows, role access matrices, and cross-module integration specifications for the AI-powered Human Resource Management System.

Version 1.0

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AI-HRMS
FUNCTIONALITY DOCUMENT

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1. Core HR & Employee Management

The Core HR module serves as the foundational layer of the AI-HRMS platform, managing the complete employee lifecycle from hire to retire. It centralizes all employee data, organizational structures, and HR administrative functions.

1.1 Features List

- Employee Master Data Management (personal, professional, emergency contacts)
- Organizational Chart & Hierarchy Management
- Job Position & Designation Management
- Employee Onboarding Checklist Automation
- Offboarding & Separation Processing
- Document Management (upload, verify, archive)
- Employee Transfer & Promotion Workflows
- Probation Period Tracking & Confirmation
- Company Policy & Handbook Distribution
- Employee Directory & Search
- Custom Fields & Dynamic Forms
- Bulk Data Import/Export (CSV, Excel)
- Audit Trail & Change Log for All Records
- AI-Powered Duplicate Employee Detection
- Automated Compliance Reminders

1.2 Configured Workflows

Workflow 1: Employee Onboarding

Step 1: HR Creates Offer **Step 2:** Employee Accepts Offer **Step 3:** Auto-Generate Employee ID **Step 4:** Provision IT Assets **Step 5:** Assign Onboarding Checklist **Step 6:** Manager Welcomes & Assigns Buddy **Step 7:** Complete Mandatory Training **Step 8:** Probation Period Starts

Workflow 2: Employee Transfer

Step 1: Manager Initiates Transfer Request **Step 2:** Current Manager Approves **Step 3:** Receiving Manager Approves **Step 4:** HR Reviews & Validates **Step 5:** Update Organizational Records **Step 6:** Reassign Reporting Lines **Step 7:** Notify Payroll for Changes **Step 8:** Transfer Effective Date Activated

Workflow 3: Employee Separation

Step 1: Initiate Resignation/Termination **Step 2:** Manager Acknowledges **Step 3:** Exit Interview Scheduled **Step 4:** Clearance Checklist Generated **Step 5:** Asset Recovery Verified **Step 6:** Final

Settlement Calculated **Step 7:** Knowledge Transfer Completed **Step 8:** Account Deactivated on Exit Date

1.3 Role Access Matrix

Feature	Super Admin	HR Admin	Payroll Specialist	Department Manager	Employee	Recruiter	L&D; Manager
Employee Data	Full	Full	Read	Team Only	Self Only	Limited	Read
Org Chart	Full	Full	Read	Read	Read	Read	Read
Transfers	Full	Full	None	Initiate	None	None	None
Offboarding	Full	Full	Read	Initiate	None	None	None
Documents	Full	Full	Read	Team Only	Self Only	None	None
Audit Trail	Full	Full	None	None	None	None	None

2. RBAC & Security System

The Role-Based Access Control module provides granular permission management across the entire HRMS platform. It enforces the principle of least privilege, ensures data security, and maintains comprehensive audit trails for compliance.

2.1 Features List

- Role Definition & Management (7 default roles, custom roles)
- Permission Assignment (module-level, feature-level, field-level)
- User Account Management & Authentication
- Multi-Factor Authentication (MFA) Enforcement
- Single Sign-On (SSO) Integration (SAML, OAuth 2.0)
- Session Management & Timeout Policies
- Data Encryption (at rest and in transit)
- IP Whitelisting & Geographic Access Controls
- API Key Management & Rate Limiting
- Security Policy Configuration (password complexity, logout)
- Comprehensive Audit Logging
- Suspicious Activity Detection & Alerts
- Data Masking for Sensitive Fields
- Role Hierarchy & Inheritance
- Temporary Access Elevation with Approval

2.2 Configured Workflows

Workflow 1: New Role Creation

Step 1: Admin Defines Role Name & Description **Step 2:** Assign Module Permissions **Step 3:** Set Feature-Level Access **Step 4:** Configure Field-Level Visibility **Step 5:** Define Role Hierarchy **Step 6:** Stakeholder Review & Approval **Step 7:** Role Activated in System

Workflow 2: Access Elevation Request

Step 1: User Requests Temporary Access **Step 2:** Manager Reviews Justification **Step 3:** Security Team Evaluates Risk **Step 4:** Time-Bound Access Granted **Step 5:** Activity Monitored During Period **Step 6:** Access Auto-Revoked at Expiry **Step 7:** Audit Entry Created

Workflow 3: Security Incident Response

Step 1: Anomaly Detected by System **Step 2:** Alert Sent to Security Team **Step 3:** Account Suspended Preemptively **Step 4:** Investigation Initiated **Step 5:** Root Cause Identified **Step 6:** Remediation Applied **Step 7:** Account Restored with Enhanced Monitoring

2.3 Role Access Matrix

Feature	Super Admin	HR Admin	Payroll Specialist	Department Manager	Employee	Recruiter	L&D; Manager
Role Mgmt	Full	Read	None	None	None	None	None
Permissions	Full	Limited	None	None	None	None	None
MFA Config	Full	None	Self	Self	Self	Self	Self
Audit Logs	Full	Read	None	Team Only	None	None	None
Security Policies	Full	None	None	None	None	None	None
API Keys	Full	None	None	None	None	None	None

3. AI-Driven Talent Acquisition & Onboarding

This module leverages artificial intelligence to streamline the entire recruitment pipeline, from job requisition to candidate onboarding. AI-powered resume screening, interview scheduling, and predictive hiring analytics reduce time-to-hire and improve quality of hire.

3.1 Features List

- Job Requisition & Approval Workflow

- AI-Powered Job Description Generator
- Multi-Channel Job Posting (internal, portals, social media)
- AI Resume Screening & Ranking (NLP-based matching)
- Candidate Pipeline Management (Kanban view)
- Automated Interview Scheduling with Calendar Integration
- Structured Interview Scorecards
- AI-Driven Candidate Assessment Tests
- Offer Letter Generation & E-Signature
- Background Verification Integration
- Pre-Onboarding Task Automation
- Onboarding Buddy Assignment System
- New Hire 30-60-90 Day Check-in
- Recruitment Analytics & Funnel Reports
- Candidate Experience Survey & NPS Tracking

3.2 Configured Workflows

Workflow 1: End-to-End Recruitment

Step 1: Hiring Manager Creates Requisition **Step 2:** HR Approves Requisition **Step 3:** AI Generates Job Description **Step 4:** Post to Multiple Channels **Step 5:** AI Screens & Ranks Applications **Step 6:** Shortlisted for Interview **Step 7:** Panel Conducts Interviews **Step 8:** Offer Extended & Accepted

Workflow 2: Interview Scheduling

Step 1: Recruiter Selects Interviewers **Step 2:** System Checks Calendar Availability **Step 3:** Automated Invite Sent to All Parties **Step 4:** Candidate Confirms Time Slot **Step 5:** Interview Link/Location Generated **Step 6:** Scorecard Template Assigned **Step 7:** Post-Interview Feedback Collected **Step 8:** Consolidated Rating Computed

Workflow 3: Pre-Onboarding Automation

Step 1: Offer Accepted **Step 2:** Background Verification Initiated **Step 3:** IT Asset Request Generated **Step 4:** System Access Provisioned **Step 5:** Welcome Kit Dispatched **Step 6:** Day-1 Schedule Created **Step 7:** Buddy Assigned **Step 8:** Employee Marked as Active

3.3 Role Access Matrix

Feature	Super Admin	HR Admin	Payroll Specialist	Department Manager	Employee	Recruiter	L&D; Manager
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Job Requisition	Full	Full	None	Initiate	None	Full	None
Resume Screening	Full	Full	None	Review	None	Full	None
Interview Mgmt	Full	Full	None	Full	None	Full	None
Offer Letters	Full	Full	Read	Approve	None	Draft	None
Onboarding	Full	Full	Read	Full	Self Only	Full	Read
Analytics	Full	Full	None	Team Only	None	Full	None

4. Time & Attendance Management

This module automates time tracking, attendance monitoring, and leave management with AI-driven anomaly detection. It integrates with biometric devices, GPS, and calendar systems for accurate workforce time data.

4.1 Features List

- Multi-Mode Attendance Capture (Biometric, GPS, Web, Mobile)
- Real-Time Attendance Dashboard
- Shift Management & Scheduling (Rotational, Fixed, Flexible)
- AI-Powered Attendance Anomaly Detection
- Leave Management (Apply, Approve, Balance Tracking)
- Leave Policy Configuration (Carry-forward, Encashment)
- Holiday Calendar Management (Multi-Region)
- Overtime Tracking & Auto-Calculation
- Comp-Off Request & Approval
- Timesheet Submission & Approval
- Project-Based Time Tracking
- Attendance Regularization Workflow
- Late Arrival / Early Departure Tracking
- AI-Driven Absenteeism Pattern Analysis
- Integration with Payroll for Salary Calculation

4.2 Configured Workflows

Workflow 1: Leave Application

Step 1: Employee Applies for Leave **Step 2:** System Validates Leave Balance **Step 3:** Auto-Route to Manager **Step 4:** Manager Approves/Rejects **Step 5:** Team Calendar Updated **Step 6:** HR Notified for Records **Step 7:** Leave Balance Deducted **Step 8:** Payroll Adjustment Triggered

Workflow 2: Attendance Regularization

Step 1: Employee Identifies Discrepancy **Step 2:** Submits Regularization Request **Step 3:** Manager Reviews & Verifies **Step 4:** HR Validates Supporting Evidence **Step 5:** Attendance Record Updated **Step 6:** Payroll Impact Assessment **Step 7:** System Log Updated **Step 8:** Anomaly Score Recalculated

Workflow 3: Shift Schedule Change

Step 1: Manager Initiates Shift Change **Step 2:** System Checks Coverage Requirements **Step 3:** Affected Employees Notified **Step 4:** Employee Acknowledges Change **Step 5:** HR Reviews Compliance **Step 6:** Shift Calendar Updated **Step 7:** Attendance Rules Adjusted **Step 8:** Payroll Rules Synced

4.3 Role Access Matrix

Feature	Super Admin	HR Admin	Payroll Specialist	Department Manager	Employee	Recruiter	L&D; Manager
Attendance	Full	Full	Read	Team Only	Self Only	None	Read
Leave Mgmt	Full	Full	Read	Approve	Self Only	None	None
Shift Mgmt	Full	Full	Read	Team Only	Read	None	None
Timesheets	Full	Full	Read	Approve	Self Only	None	None
Anomaly Detection	Full	Read	None	Team Only	None	None	None
Reports	Full	Full	Read	Team Only	None	None	None

5. Payroll & Expense Management

The Payroll module delivers end-to-end salary processing with automated tax calculations, statutory compliance, and multi-country payroll support. The Expense module streamlines claim submission, approval, and reimbursement with policy-based validation.

5.1 Features List

- Automated Salary Calculation (Basic, HRA, Allowances, Deductions)
- Tax Computation & TDS Management (Multi-Country)
- Statutory Compliance (PF, ESI, PT, LWF)
- Bank Integration for Salary Disbursement
- Payslip Generation & Distribution (PDF, Email)
- Salary Structure Configuration & Versioning
- Investment Declaration & Proof Collection
- Form 16 / Tax Certificate Generation

- Expense Claim Submission with Receipt Upload
- Multi-Level Expense Approval Workflow
- Policy-Based Expense Auto-Validation
- Mileage & Travel Expense Calculator
- Corporate Card Integration & Reconciliation
- Advance Salary & Loan Management
- Payroll Analytics & Cost Distribution Reports

5.2 Configured Workflows

Workflow 1: Monthly Payroll Processing

Step 1: Payroll Admin Initiates Run **Step 2:** System Pulls Attendance & Leave Data **Step 3:** OT & Deductions Calculated **Step 4:** Tax & Statutory Computed **Step 5:** Draft Paysheets Generated **Step 6:** Manager Review & Approval **Step 7:** Bank File Created & Uploaded **Step 8:** Payslips Distributed to Employees

Workflow 2: Expense Claim Processing

Step 1: Employee Submits Claim with Receipts **Step 2:** AI Validates Against Policy **Step 3:** Auto-Route to Manager **Step 4:** Manager Approves/Queries **Step 5:** Finance Team Reviews **Step 6:** Reimbursement Scheduled **Step 7:** Payment Processed **Step 8:** Employee Notified of Settlement

Workflow 3: Investment Declaration

Step 1: Employee Declares Investments **Step 2:** System Validates Limits & Rules **Step 3:** Manager Reviews if Required **Step 4:** HR Acknowledges Declaration **Step 5:** TDS Adjusted for Remaining Months **Step 6:** Proof Collection Deadline Set **Step 7:** Proofs Submitted & Verified **Step 8:** Final Tax Computation Updated

5.3 Role Access Matrix

Feature	Super Admin	HR Admin	Payroll Specialist	Department Manager	Employee	Recruiter	L&D; Manager
Salary Processing	Full	Full	Full	None	None	None	None
Tax Mgmt	Full	Read	Full	None	Self Only	None	None
Expense Claims	Full	Full	Process	Approve	Self Only	Self Only	Self Only
Payslips	Full	Full	Full	None	Self Only	Self Only	Self Only
Statutory	Full	Read	Full	None	Self Only	None	None
Analytics	Full	Full	Full	Team Cost Only	None	None	None

6. Performance & Talent Development

This module enables continuous performance management through configurable review cycles, 360-degree feedback, goal alignment (OKRs/KPIs), and AI-driven talent identification for succession planning and career development.

6.1 Features List

- Configurable Performance Review Cycles (Annual, Bi-Annual, Quarterly)
- Goal Setting & Tracking (OKRs, KPIs, MBOs)
- Cascading Goals (Company > Department > Individual)
- 360-Degree Feedback Collection
- Continuous Feedback & Check-ins
- Performance Rating Scales (Customizable)
- AI-Powered Performance Bias Detection
- Talent Matrix / 9-Box Grid
- Succession Planning & Pipeline
- Career Path & Development Plans
- Competency Framework & Assessment
- Calibration Sessions for Rating Normalization
- PIP (Performance Improvement Plan) Management
- Promotion & Increment Recommendation Engine
- Performance Analytics & Trend Reports

6.2 Configured Workflows

Workflow 1: Annual Performance Review

Step 1: HR Initiates Review Cycle **Step 2:** Employee Completes Self-Assessment **Step 3:** Manager Rates & Reviews **Step 4:** 360 Feedback Collected **Step 5:** Calibration Session Held **Step 6:** Final Rating Assigned **Step 7:** Employee Acknowledges **Step 8:** Development Plan Created

Workflow 2: PIP (Performance Improvement Plan)

Step 1: Manager Identifies Low Performance **Step 2:** HR Validates & Reviews **Step 3:** PIP Created with Milestones **Step 4:** Employee Acknowledges PIP **Step 5:** Regular Check-ins Scheduled **Step 6:** Progress Assessed at Checkpoints **Step 7:** PIP Completed - Pass/Fail **Step 8:** Next Action (Recovery/Separation)

Workflow 3: Succession Planning

Step 1: Identify Critical Positions **Step 2:** Define Competency Requirements **Step 3:** AI Ranks Potential Successors **Step 4:** HR Reviews Shortlist **Step 5:** Development Plans Assigned **Step 6:** Readiness Assessment Conducted **Step 7:** Succession Map Updated **Step 8:** Board Review & Approval

6.3 Role Access Matrix

Feature	Super Admin	HR Admin	Payroll Specialist	Department Manager	Employee	Recruiter	L&D; Manager
Goal Setting	Full	Full	None	Team Only	Self Only	None	Read
Reviews	Full	Full	None	Team Only	Self Only	None	Read
360 Feedback	Full	Full	None	Provide	Provide	None	Read
Talent Matrix	Full	Full	None	Team Only	None	None	Read
Succession	Full	Full	None	Initiate	None	None	Read
PIP	Full	Full	None	Initiate	Self Only	None	Support

7. Learning & Development

The L&D; module provides a comprehensive learning management platform with AI-curated learning paths, course management, certification tracking, and skills gap analysis to build a future-ready workforce.

7.1 Features List

- Learning Management System (LMS) with Course Catalog
- AI-Powered Learning Path Recommendations
- Course Creation & Content Management (SCORM, Video, Docs)
- Assessment & Quiz Engine
- Certification Management & Tracking
- Skills Gap Analysis (AI-Driven)
- Training Budget Allocation & Tracking
- External Training Request & Approval
- Compliance Training Automation
- Gamification (Badges, Points, Leaderboards)
- Instructor-Led Training (ILT) Scheduling
- Virtual Classroom Integration
- Learning Analytics & ROI Measurement
- Mentorship Program Management

7.2 Configured Workflows

Workflow 1: Training Nomination

Step 1: L&D; Publishes Training Calendar **Step 2:** Manager Nominates Employee **Step 3:** Employee Accepts Nomination **Step 4:** Prerequisites Checked **Step 5:** Enrollment Confirmed **Step 6:** Training Delivered (Online/ILT) **Step 7:** Assessment Completed **Step 8:** Certificate Issued & Skills Updated

Workflow 2: External Training Request

Step 1: Employee Submits Request with Details **Step 2:** Manager Reviews & Approves **Step 3:** L&D; Validates Budget & Relevance **Step 4:** Finance Approves Budget **Step 5:** Vendor Coordination **Step 6:** Employee Attends Training **Step 7:** Knowledge Sharing Session Conducted **Step 8:** Skills Profile Updated

Workflow 3: Compliance Training Cycle

Step 1: System Identifies Due Compliance Trainings **Step 2:** Auto-Enroll Affected Employees **Step 3:** Deadline Notifications Sent **Step 4:** Employee Completes Training **Step 5:** Assessment Score Evaluated **Step 6:** Non-Compliance Escalated to Manager **Step 7:** HR Compliance Dashboard Updated **Step 8:** Audit Report Generated

7.3 Role Access Matrix

Feature	Super Admin	HR Admin	Payroll Specialist	Department Manager	Employee	Recruiter	L&D; Manager
Course Mgmt	Full	Full	None	Nominate	Enroll	None	Full
Learning Paths	Full	Full	None	Team Only	Self Only	None	Full
Certifications	Full	Full	None	Team Only	Self Only	Read	Full
Budget	Full	Read	Read	None	None	None	Full
Analytics	Full	Full	None	Team Only	Self Only	None	Full
Compliance	Full	Full	None	Team Only	Self Only	None	Full

8. Analytics & Reporting

The Analytics module provides data-driven insights across all HR functions with interactive dashboards, configurable reports, and AI-powered predictive analytics for workforce planning and strategic decision-making.

8.1 Features List

- Executive HR Dashboard (Real-Time KPIs)
- Headcount & Workforce Demographics Analytics
- Attrition Analysis & Prediction (AI-Driven)
- Recruitment Funnel & Time-to-Hire Reports
- Attendance & Leave Pattern Analytics
- Payroll Cost Distribution & Trend Reports
- Performance Distribution & Calibration Analytics
- Training Effectiveness & ROI Reports
- Diversity, Equity & Inclusion (DEI) Metrics
- Custom Report Builder (Drag & Drop)
- Scheduled Report Generation & Email Distribution
- Data Export (Excel, CSV, PDF, API)
- AI-Powered Anomaly Detection in HR Data
- Benchmarking Against Industry Standards
- Predictive Workforce Planning Models

8.2 Configured Workflows

Workflow 1: Scheduled Report Generation

Step 1: Admin Configures Report Parameters **Step 2:** Set Schedule (Daily/Weekly/Monthly) **Step 3:** Define Recipient List **Step 4:** System Generates Report at Schedule **Step 5:** AI Quality Check on Data **Step 6:** Report Emailed to Recipients **Step 7:** Archived in Report Repository **Step 8:** Usage Tracking Logged

Workflow 2: Attrition Prediction

Step 1: AI Model Ingests Employee Data **Step 2:** Risk Scores Calculated per Employee **Step 3:** High-Risk Profiles Flagged **Step 4:** Manager Notified with Insights **Step 5:** Retention Action Plan Suggested **Step 6:** Manager Initiates Intervention **Step 7:** Follow-Up Tracking Activated **Step 8:** Model Retrained with Outcomes

Workflow 3: Custom Report Creation

Step 1: User Selects Data Source **Step 2:** Choose Dimensions & Metrics **Step 3:** Apply Filters & Date Range **Step 4:** Preview Report Output **Step 5:** Configure Visualization Type **Step 6:** Save Report Template **Step 7:** Set Sharing Permissions **Step 8:** Publish or Schedule

8.3 Role Access Matrix

Feature	Super Admin	HR Admin	Payroll Specialist	Department Manager	Employee	Recruiter	L&D; Manager
Dashboards	Full	Full	Payroll Only	Team Only	None	Recruitment Only	L&D; Only
Custom Reports	Full	Full	Payroll Only	Team Only	None	Recruitment Only	L&D; Only
Predictive	Full	Full	None	Team Only	None	None	None
Export	Full	Full	Payroll Only	Team Only	None	Recruitment Only	L&D; Only
Scheduling	Full	Full	Payroll Only	Team Only	None	Recruitment Only	L&D; Only
Benchmarking	Full	Full	None	None	None	None	None

9. Employee Self-Service & Collaboration

The ESS module empowers employees with self-service capabilities for common HR tasks, while the collaboration features enable seamless communication, team coordination, and knowledge sharing across the organization.

9.1 Features List

- Employee Self-Service Portal (Profile, Documents, Requests)
- Manager Self-Service (Team Actions, Approvals)
- Company Announcement Board
- Team Calendar & Event Management
- Employee Directory with Org Chart Navigation
- Help Desk & Ticket System (HR Queries)
- AI Chatbot for HR Policy Queries
- Document Request & Verification
- Service Certificate Generation
- Peer Recognition & Rewards Program
- Survey & Poll Management
- Discussion Forums & Knowledge Sharing

- Meeting Room Booking System
- Mobile App with Full ESS Features
- Push Notifications & Communication Hub

9.2 Configured Workflows

Workflow 1: HR Help Desk Ticket

Step 1: Employee Submits Ticket **Step 2:** AI Categorizes & Routes **Step 3:** Assigned to HR Team Member **Step 4:** Agent Investigates Issue **Step 5:** Resolution Proposed **Step 6:** Employee Confirms/Rejects **Step 7:** Ticket Closed & Rated **Step 8:** Knowledge Base Updated

Workflow 2: Peer Recognition

Step 1: Employee Nominates Peer **Step 2:** Selects Recognition Category **Step 3:** Adds Appreciation Message **Step 4:** Manager Endorses (Optional) **Step 5:** Recognition Posted on Wall **Step 6:** Points/Credits Awarded **Step 7:** Monthly Leaderboard Updated **Step 8:** Quarterly Rewards Processed

Workflow 3: Document Request

Step 1: Employee Requests Document **Step 2:** System Validates Eligibility **Step 3:** Auto-Generated if Template-Based **Step 4:** HR Reviews if Manual Required **Step 5:** Manager Approval if Needed **Step 6:** Document Prepared & Uploaded **Step 7:** Employee Downloads Document **Step 8:** Request Archived

9.3 Role Access Matrix

Feature	Super Admin	HR Admin	Payroll Specialist	Department Manager	Employee	Recruiter	L&D; Manager
Self-Service	Full	Full	Self Only	Team Only	Self Only	Self Only	Self Only
Announcements	Full	Full	Read	Create	Read	Read	Create
Help Desk	Full	Full	Self Only	Team Only	Self Only	Self Only	Self Only
Recognition	Full	Full	Participate	Endorse	Participate	Participate	Participate
Directory	Full	Full	Read	Full	Read	Read	Read
Surveys	Full	Full	Participate	Create	Participate	Participate	Create

10. Cross-Module Workflows

Cross-module workflows span multiple HRMS modules to enable end-to-end business processes that require coordination across different functional areas. These workflows ensure data consistency, eliminate manual handoffs, and provide unified employee experiences.

10.1 Employee Lifecycle Workflow

This workflow tracks the complete journey of an employee from candidate to alumni, spanning Modules 3 (Recruitment), 1 (Core HR), 4 (Attendance), 5 (Payroll), 6 (Performance), 7 (L&D;), and 9 (ESS).

Complete Employee Lifecycle

Step 1: Recruitment (Module 3) **Step 2:** Onboarding (Module 1) **Step 3:** Probation Confirmation (Module 1) **Step 4:** Attendance Tracking (Module 4) **Step 5:** Performance Reviews (Module 6) **Step 6:** Learning & Development (Module 7) **Step 7:** Payroll Processing (Module 5) **Step 8:** Separation & Alumni (Module 1)

10.2 Approval Hierarchy Workflow

A unified approval framework that applies across all modules for consistent governance. Any request (leave, expense, transfer, training) follows this standardized hierarchy.

Unified Approval Hierarchy

Step 1: Employee Initiates Request **Step 2:** System Validates Against Policy (AI) **Step 3:** Auto-Approve if Within Threshold **Step 4:** Route to Direct Manager **Step 5:** Escalate to Department Head if Exceeds Limit **Step 6:** HR Review for Compliance **Step 7:** Finance Approval if Budget Impact **Step 8:** Action Executed & Stakeholders Notified

10.3 Notification System Workflow

The notification system provides real-time, context-aware communications across all modules, ensuring stakeholders are informed at every critical juncture.

Notification Dispatch

Step 1: Event Triggered in Any Module **Step 2:** Notification Engine Evaluates Rules **Step 3:** Determine Recipients by Role & Context **Step 4:** Select Channel (Email/Push/SMS/In-App) **Step 5:** Personalize Message Content **Step 6:** Schedule Based on Urgency **Step 7:** Deliver Notification **Step 8:** Track Delivery & Read Receipts

11. Complete Role-Module Access Matrix

The following matrix provides a consolidated view of access levels for all 7 roles across all 9 modules. Access levels are: **Full** (complete read/write/admin), **Read** (view only), **Limited** (role-specific partial access), **None** (no access).

Module	Super Admin	HR Admin	Payroll Specialist	Dept. Manager	Employee	Recruiter	L&D; Manager
Core HR	Full	Full	Read	Limited	Self	Limited	Read
RBAC & Security	Full	Limited	None	None	Self	None	None
Talent Acquisition	Full	Full	None	Limited	None	Full	Read
Time & Attendance	Full	Full	Read	Limited	Self	None	Read
Payroll & Expense	Full	Full	Full	Limited	Self	Self	Self
Performance & Talent	Full	Full	None	Limited	Self	None	Read
Learning & Development	Full	Full	None	Limited	Self	Read	Full
Analytics & Reporting	Full	Full	Limited	Limited	None	Limited	Limited
ESS & Collaboration	Full	Full	Self	Limited	Self	Self	Self

Table 1: Complete Role-Module Access Matrix (7 Roles x 9 Modules)

12. Workflow Configuration Guide

This guide provides administrators with the framework for configuring, customizing, and managing workflows across the AI-HRMS platform.

12.1 Workflow Configuration Principles

- **Principle of Least Privilege:** Workflows should only grant access and trigger actions necessary for the specific business process.
- **Fail-Safe Design:** Every workflow must have timeout handling, escalation paths, and rollback mechanisms.
- **Audit-First Approach:** All workflow state transitions must be logged with timestamp, actor, and reason.

- **AI-Assisted Routing:** Leverage the AI engine for intelligent routing, policy validation, and anomaly detection at each workflow step.
- **Modular Composition:** Complex workflows should be composed from reusable sub-workflows for maintainability.

12.2 Workflow Configuration Parameters

Parameter	Description	Example
Trigger Event	The event that initiates the workflow	Employee submits leave request
Participants	Roles/users involved in the workflow	Employee, Manager, HR Admin
Steps	Sequential actions in the workflow	Submit > Validate > Approve > Execute
Conditions	Business rules evaluated at each step	Leave balance > 0, Probation completed
Timeout	Maximum time before escalation	48 hours for manager approval
Escalation	Action when timeout is reached	Auto-escalate to Department Head
Notifications	Alerts sent at each transition	Email to manager, Push to employee
Rollback	Actions if workflow is rejected	Restore leave balance, Notify employee
Audit Fields	Data captured for compliance	Approver, Timestamp, Comments
AI Integration	AI features within the workflow	Policy validation, Anomaly detection

Table 2: Workflow Configuration Parameters

12.3 Workflow Lifecycle Management

Workflow Lifecycle

Step 1: Define Requirements **Step 2:** Design Workflow Steps & Rules **Step 3:** Configure in Admin Panel **Step 4:** Test with Sample Data **Step 5:** Stakeholder Approval **Step 6:** Deploy to Production **Step 7:** Monitor Performance & Bottlenecks **Step 8:** Iterate & Optimize

12.4 Best Practices

- Always start with a standard template before customizing workflows.
- Keep approval chains short - ideally no more than 3 levels.
- Use AI auto-approval for low-risk, policy-compliant requests to reduce bottlenecks.

- Set up monitoring dashboards for workflow completion rates and average processing times.
- Conduct quarterly workflow audits to identify unused or redundant steps.
- Document all custom workflows with clear business justification.
- Test all workflow changes in a sandbox environment before production deployment.
- Ensure all workflows have defined SLAs for each approval step.

12.5 Troubleshooting Common Issues

Issue	Root Cause	Resolution
Workflow stuck at approval step	Approver inactive or unassigned	Set auto-escalation timeout; verify approver mapping
Incorrect data passed between steps	Field mapping misconfiguration	Validate field bindings in workflow designer
Duplicate notifications sent	Multiple trigger rules active	Audit notification rules; consolidate duplicates
AI validation producing false rejections	Overly strict policy rules	Review AI model confidence thresholds; add exceptions
Workflow not visible to user	Role permission gap	Check RBAC module for workflow access assignment

Table 3: Common Workflow Issues and Resolutions